SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Computer Vision with OpenCV **COURSE CODE: DSA02**

COURSE OUTCOME COVERED
CO2: *Apply image processing and feature extraction techniques, including edge detection, motion estimation, and optical flow, for analyzing images.. (BL3)*

Assessment Tool Weightage: 20%

QUESTIONS: 2 **Total Marks:20**

Annexure A

Class Test

Code of Conduct:

I Peram.Mahitha(192524136)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *P.Mahitha*

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions	
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation	
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis	
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers	
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps	
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand	

Annexure A

Class Test

Question 1: Image Preprocessing and Feature Extraction

Scenario: A drone is used to capture aerial images of a farmland to identify pest-affected crop regions. Due to sensor limitations, wind movement, and changing sunlight, the captured images contain noise, shadows, and uneven illumination. Pest-affected areas usually show visible texture and color variations compared to healthy crop regions.

Answer the following questions:

a) Preprocessing Pipeline — 4 Marks

Propose a suitable preprocessing pipeline to improve the quality of the farmland images before analysis. Your answer should include methods for noise reduction, illumination correction, contrast enhancement, and normalization. Justify the purpose of each step.

b) Feature Extraction — 3 Marks

Choose an appropriate feature extraction method to identify texture changes caused by pest damage. Explain why the selected method is suitable for detecting affected crop regions.

c) Rotation and Scale Variations — 3 Marks

Drone images may be captured from different heights and angles. Discuss how rotation and scale variations can affect feature detection. Suggest suitable methods to make the feature extraction process more robust.

Ans)

Image Preprocessing and Feature Extraction

a) Preprocessing Pipeline

- **Noise Reduction (Median Filtering):**
    - **Purpose:** Removes high-frequency salt-and-pepper noise caused by electronic sensor limitations.
    - **Justification:** Median filtering replaces pixels with the median of neighborhood pixels. This removes noise while preserving sharp boundaries between healthy and pest-affected crop edges**.**
- **Illumination Correction (Homomorphic Filtering):**
    - **Purpose:** Corrects uneven lighting and shadows caused by cloud cover or shifting sunlight.
    - **Justification:** It operates in the frequency domain to separate low-frequency illumination from high-frequency reflectance. This dampens the illumination component to reveal underlying crop details hidden in shadows**.**
- **Contrast Enhancement (CLAHE):**
    - **Purpose:** Enhances regional color and texture differences without over-amplifying noise.
    - **Justification:** Contrast Limited Adaptive Histogram Equalization (CLAHE) operates on small, local image tiles. This prevents global over-saturation while bringing out subtle variations in pest-damaged leaves.
- **Normalization (Min-Max Scaling):**
    - **Purpose:** Rescales pixel intensity values to a standard range, such as [0, 1]or [0, 255].

- **Justification:** Normalization ensures that variations in global exposure do not bias downstream machine learning classifiers or feature extractors.

b) Feature Extraction

Selected Method: Local Binary Patterns (LBP) combined with Color Histograms (HSV space).

- **Texture Suitability:** LBP computes local contrast by comparing each pixel to its neighbors. Pest damage degrades leaf structure, creating rough, irregular textures. LBP forms a robust binary descriptor that captures these exact micro-textural changes.
- **Color Suitability:** Color histograms in the HSV (Hue, Saturation, Value) space isolate the actual color information (Hue) from lighting changes (Value). Pest-affected regions often turn yellow or brown, deviating from healthy green vegetation.
- **Combined Strength:** Fusing LBP and HSV histograms builds a joint feature vector. This vector uniquely identifies damaged zones based on both leaf decay patterns and discolouration.

c) Rotation and Scale Variations

Impact on Feature Detection

- **Scale Variations:** Images taken from higher altitudes make crops appear smaller. Standard pixel-sized texture windows will miss patterns because the relative scale of the leaf damage shrinks.
- **Rotation Variations:** Changes in drone flight heading rotate the texture grid. Non-invariant descriptors will generate entirely different feature vectors for the exact same patch of land.

Robust Methods

- **Scale Invariance:** Use Scale-Invariant Feature Transform (SIFT) or Speeded-Up Robust Features (SURF). These methods utilize a scale-space pyramid to detect stable keypoints regardless of altitude**.**
- **Rotation Invariance:** Apply Rotation-Invariant Local Binary Patterns (LBP^{ri}). This technique circularly shifts the binary code to its minimum possible value, ensuring uniform descriptors regardless of camera angle

Question 2: Stereo Vision and Motion Estimation

Scenario: A self-driving car is equipped with stereo cameras to navigate through a busy street. The system must estimate the distance of nearby vehicles and pedestrians in real time. It should also track moving objects accurately to avoid collisions, especially when pedestrians or vehicles change speed suddenly.

Answer the following questions:

a) Stereo Disparity and Depth Estimation — 4 Marks

Explain how stereo cameras use the difference between left and right image views to estimate the depth of objects. Discuss how disparity changes for nearby and distant objects.

b) Optical Flow for Motion Tracking — 3 Marks

If a pedestrian suddenly changes speed or direction, explain how optical flow helps in tracking the pedestrian's motion between consecutive video frames.

c) Errors and Mitigation Techniques — 3 Marks

Identify possible sources of error in depth estimation and motion tracking, such as camera misalignment, poor lighting, occlusion, fast motion, or textureless regions. Suggest suitable techniques to reduce these errors.

Ans)

Stereo Vision and Motion Estimation

a) Stereo Disparity and Depth Estimation

Stereo vision uses two horizontally separated cameras to capture slightly different views of the same scene. Depth is calculated by finding corresponding points in both images and measuring their **disparity** (the horizontal pixel shift between the two points).

The mathematical relationship is defined by the triangulation formula:

$$\text{Depth}(Z)= (fXB)/d$$

f: Focal length of the camera lenses.

- B: Baseline distance (horizontal separation between the two cameras).
- d: Disparity ($x_{left} - x_{right}$).

Disparity Behavior

- **Nearby Objects:** Pedestrians or cars close to the vehicle exhibit a **high disparity**. They shift drastically between the left and right frames, allowing for highly accurate, high-resolution depth calculations.
- **Distant Objects:** Faraway items display a **low disparity**, changing very little between frames. As distance increases, disparity approaches zero, making depth estimation less precise for faraway backgrounds.

b) Optical Flow for Motion Tracking

Optical flow tracks motion by calculating the apparent velocity vectors of pixels between consecutive video frames. It relies on the brightness constancy assumption, meaning a tracked point's intensity remains relatively constant over time.

$\frac{\partial I}{\partial x}u+\frac{\partial I}{\partial y}v+\frac{\partial I}{\partial t}=0$

Error Source	Impact on System	Mitigation Technique
Camera Misalignment	Skews the epipolar lines, corrupting the baseline geometry and causing incorrect disparity math.	**Online Geometric Calibration:** Run automated background rectification algorithms that continuously align pixel rows using static horizon points.
Textureless Regions	Smooth walls or uniform roads lack unique pixel markers, preventing the stereo matcher from finding corresponding points.	**Semi-Global Matching (SGM):** Incorporate structural path constraints from surrounding areas to infer depth in blank zones.
Fast Motion & Occlusion	Causes motion blur and breaks pixel tracking when objects temporarily hide behind one another.	**Kalman Filtering:** Fuse optical flow vectors with a constant-velocity kinematic model to intelligently estimate object paths during blind spots or blurred frames.